

YIQI LIANG

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EDUCATION

■ The Hong Kong University of Science and Technology

Guangzhou, China

MPhil in Computational Media and Arts, Information Hub

Sep. 2023 - Oct. 2025

- Supervised by *Prof. Mingming Fan* (Primary) and *Prof. Yuyu Luo* (Co)
- GPA: 3.94/4.30

■ Northeastern University

Hebei, China

B.Eng in Computer Science and Technology

Sep. 2018 - Jul. 2022

- GPA: 89.4/100 (Ranking Top 7.8%)

RESEARCH INTEREST

Visual Generation, Multi-Agent Systems, Human-AI Interaction

PUBLICATIONS AND MANUSCRIPTS

■ Ongoing Work

[O1] **Yiqi Liang**, Xiaodong Cun. *SketchClaw: Agentic Multi-Concept Scene Sketching*. Ongoing. 2026.

■ First-Author Publications

[C1] **Yiqi Liang**, Ying Liu, Dandan Long and Ruihui Li. *PortraVec: Image-Based Portrait Vectorization with Text-Guided Manipulation*. *ICME*. 2026. **Accepted**

[C2] **Yiqi Liang**, Ying Liu, Dandan Long and Ruihui Li. *MROSS: Multi-Round Region-based Optimization for Scene Sketching*. *ICME*. 2025. **Accepted**

■ Second/Third-Author Publications

[J1] Fan Lin, **Yiqi Liang**, Qianjie Wei, Mingshuo Li, Chutian Jiang and Mingming Fan. *VR-Shuttlecock: Exploring the Possibility of Virtual Reality (VR) Shuttlecock Kicking with Multi-Sensory Feedback for Empowering Older Adults in Balance Training*. *IMWUT*. 2025. **Accepted**

[C3] Qianjie Wei, Xiaoying Wei, **Yiqi Liang**, Nuonan Si and Mingming Fan. *RemoteChess: Enhancing Older Adults' Social Connectedness via Designing a Virtual Reality Chinese Chess (Xiangqi) Community*. *CHI*. 2025. **Accepted**

[C4] Junxian Chen#, Ying Liu#, **Yiqi Liang**, Dandan Long and Ruihui Li. *SD-Net: Spatially-Disentangled Point Cloud Completion Network*. *ACM MM*. 2023. **Accepted**

Equal contribution.

SELECTED PROJECTS

■ Research Assistant

Dec. 2025 - Sep. 2026

Generative & Intelligent Visual Computing (GVC) Lab, GBU, Supervised by Prof. Xiaodong Cun

• [O1] Agentic Multi-Concept Scene Sketching.

Developed an agentic framework for editable multi-concept scene sketch generation, where agents perform scene decomposition, spatial planning, execution, and multimodal verification through an iterative Plan-Verify-Correct loop. Introduced short-term failure memory and long-term experience memory to improve layout reasoning and iterative correction. And incorporated a video model prior for structure-aware vector refinement while preserving object structure and spatial composition. The framework achieves improved semantic alignment, spatial composition, and object completeness over representative vector sketch generation baselines.

- **Enabling Co-Creative Visual Storytelling for Older Adults with Generative AI**

Investigated the challenges and needs of older adults during visual story preparation, including memory retrieval, visual content organization, and expressive representation. Designed, implemented, and evaluated StoryMosaic, a GenAI-powered interactive system that scaffolds memory elicitation, visual content generation and composition, and expressive enhancement to support more effective and controllable autobiographical storytelling.

- **From Scenarios to Strategies: A Systematic Framework for Understanding Spatial Information Needs of Blind and Low Vision people**

Investigated prior works and practices regarding the spatial information needs of blind and low vision (BLV) people in specific scenarios, such as navigation. And designed a framework to systematically capture the spatial information needs of BLV people, aims to reveal neglected spatial information needs, identify new research directions, and guide design improvements of assistive tools.

- **LayerTouch: Layer-Based Image Exploration for Blind and Low Vision Users**

Designed and developed LayerTouch, an accessible image exploration tool that enables blind and low vision (BLV) users to interact with layered visual content on touchscreens. The tool allows users to navigate image objects by depth, providing structured access to foreground and background elements, enhancing image understanding and grants users a stronger sense of agency.

- **Assisting Medication Information Leaflets Reading for Older Adults: Current Challenges and the Explorations of MediSUM**

Investigated the practices and challenges faced by Chinese older adults in reading medication leaflets. Designed and implemented a WeChat Mini Program with AI-assisted functions to assist older adults in reading and understanding medication leaflets, improving medication safety.

■ Research Assistant

Sep. 2022 - Aug. 2023

ShapeLab, Hunan University, Supervised by Prof. Ruihui Li

- **[C1] Image-Based Portrait Vectorization with Text-Guided Manipulation.**

Developed a portrait vectorization framework to convert portrait images into editable vector sketches with text-guided manipulation. Also proposed Attention-aware Offset Sampling to preserve facial details and Region-based Parameter Freezing for localized editing while maintaining global consistency.

- **[C2] Multi-Round Region-based Optimization for Scene Sketching**

Developed a multi-round optimization mechanism to seamlessly integrate the sketch strokes of different regions to generate a scene sketch. Also proposed an additional stroke initialization method to ensure the completeness of the scene and the convergence of the optimization.

AWARDS & HONORS

2018/2019/2020/2021 Outstanding Student Scholarship of Northeastern University	2018 - 2022
Outstanding Graduates Prize of Northeastern University	2022
Best Code Award (2/269, 0.74%), HUAWEI Cloud BlockChain College Competition	2021
First prize (5%), The 14th Hebei Province Programming Contest	2021

SKILLS & LANGUAGES

Languages Proficiency: Mandarin (Native), English (Fluent, IELTS 6.5)

Programming Languages: Python, C++, C, C#, SQL, JavaScript, p5.js, HTML/CSS,

Technologies & Tools: Vision-Language Models, Generative Model, Vector Graphics and Visual Computing, VR/AR/XR.

Others: Skilled in Figma, Adobe kits Adobe Kit (Photoshop, Illustrator)